

Summary

Ph.D. student in Computer Science at Stony Brook University with graduate training in theoretical physics and quantum information. My research covers quantum circuit verification and equivalence checking, quantum channel learning, entanglement and nonlocality certification in quantum networks, and open quantum systems. I build Python/C++ research software and benchmark pipelines, use linear and semidefinite programming in numerical work, and have run circuit experiments on IBM Quantum devices.

Research interests Quantum computing, quantum learning theory, quantum networking, quantum information theory, quantum programming languages and verification

Education

09/2023 – Ph.D. in Computer Science

Present *Stony Brook University, New York, United States*

Advisors: Nengkun Yu, Aruna Balasubramanian

Full-time Ph.D. student. Fourth year begins Fall 2026.

09/2022 – Master of Science in Quantum Information

08/2023 *Quantum Information Center (QICS), Sorbonne Université, Paris, France*

Thesis: “NONLOCALITY DISTILLATION”

Advisors: Jean-Daniel Bancal, Mirjam Weilenmann, Peter Brown

Courses: Quantum information theory, quantum cryptography, quantum algorithms, quantum optics, photonic quantum computing, atomic physics for quantum information, quantum optomechanics

09/2021 – Master of Science in Theoretical Physics

08/2022 *International Centre for Fundamental Physics (ICFP), École normale supérieure (ENS) de Paris, France*

Thesis: “CLASSICAL SIMULATION OF SHALLOW RANDOM QUANTUM CIRCUITS”

Advisor: Omar Fawzi

Courses: Statistical physics, statistical field theory, quantum field theory, Yang–Mills theory, general relativity, quantum information theory, quantum metrology, introduction to AdS/CFT, Lie groups, Lie algebras and representations

09/2018 – Engineering Degree (Ingénieur polytechnicien)

08/2021 *École Polytechnique, Institut Polytechnique de Paris, Palaiseau, France*

Specialization: Fundamental Physics, third-year program “From Particles to Stars”

Courses: Elementary particle physics, stellar astrophysics, general relativity, perturbative quantum field theory, algebra, Galois theory, Lie groups, Lie algebras and representations, optimization and control of dynamic systems

09/2015 – Bachelor of Science, Gifted Program in Physics

06/2018 *Hanoi University of Science, Hanoi, Vietnam*

09/2012 – HUS High School for Gifted Students, Mathematics

05/2015 *Hanoi University of Science, Hanoi, Vietnam*

Research Experience

09/2023 – Research Assistant

Present *Department of Computer Science, Stony Brook University*

Advisors: *Nengkun Yu, Aruna Balasubramanian*

Quantum circuit verification

- Develop Python/C++ software for scalable equivalence checking and verification of shallow quantum circuits, using symbolic methods, local-projection fingerprints, weighted binary decision diagrams, and simulation-based validation.
- Write and maintain the C++ and Python code of MANJUSHRI, including core data structures, the command-line interface, the build system, and the benchmark harness.
- Build benchmark suites over structured and randomized Clifford+ T circuits, with parameter sweeps, job scripts, structured logging, result aggregation, plotting, and runtime and failure-mode analysis.
- Debug correctness and performance issues with `gdb`, `valgrind`, and profiling tools. Read and modify third-party research code (QUASIMODO, Qiskit) to integrate and extend it.

Quantum learning and noise

- Study quantum channel learning and Pauli-noise characterization, including sample-complexity analysis and validation of estimators against simulation.
- Run and analyze circuit experiments on IBM Quantum cloud devices through group projects, comparing hardware data with simulator expectations and circuit-level noise models.
- Study open quantum systems, including dephasing in bosonic and fermionic fields and the emergence of classical stochasticity.

Quantum networks

- Certify entanglement between network nodes from measured CHSH violation (QCE 2026, lead author).
- Contributed to the design and analysis of a verification-based quantum transmission protocol (ICDCS 2026, Distinguished Paper Award).
- Study how entanglement, steering, and nonlocality evolve under noise and during entanglement swapping, including limitations of generalized entanglement swapping under noise (in preparation).

Lab computing

- Set up and maintained workstations and shared Linux servers for the group, including software installation, environment configuration, user accounts and permissions, and routine updates.
- Wrote setup scripts and documentation so new lab members could install tools and connect to shared resources on their own.

05/2023 – Research Internship

09/2023 *Center for Theoretical Physics (IPhT), Commissariat à l'Énergie Atomique (CEA) Saclay, France*

Advisors: *Jean-Daniel Bancal, Mirjam Weilenmann, Peter Brown*

Project: “NONLOCALITY DISTILLATION”

- Formulated quantum-information constraints as linear programs and developed optimization-based methods for nonlocality distillation, with applications to quantum communication and quantum key distribution.
- Wrote Python code for numerical optimization experiments, ran batches of jobs on shared infrastructure, compared protocol variants, and reported results in written and oral updates.
- Worked with an international team in English and French.

04/2022 – Research Internship

07/2022 *Centre Inria de Lyon, Laboratoire de l'Informatique du Parallélisme (LIP), ENS de Lyon, France*

Advisor: *Omar Fawzi*

Project: “CLASSICAL SIMULATION OF SHALLOW RANDOM QUANTUM CIRCUITS”

- Studied existing techniques for classical simulation of shallow random quantum circuits.
- Developed and tested semidefinite-programming-based simulation methods, contributed C++/Python code, designed test cases, and evaluated behavior across circuit families.

03/2021 – Research Internship

07/2021 *Irène Joliot-Curie Laboratory of Physics of Two Infinities (IJCLab), Université Paris-Saclay, Orsay, France*

Advisors: *Louis Fayard, Zhiqing Zhang*

Project: “PDF UNCERTAINTIES AND THE W BOSON MASS MEASUREMENT”

- Analyzed how parton distribution function uncertainties contribute to the systematic uncertainty of the W boson mass measurement in the ATLAS experiment at CERN.
- Processed and analyzed multi-gigabyte ATLAS datasets with ROOT and Python, propagated uncertainties, and presented results to senior scientists.

06/2020 – Research Internship

08/2020 *CEMAFROID SAS, Fresnes, France*

Project: “CONFINEMENT OF A REFRIGERATED BOX USING LATERAL AIR CURTAINS”

Designed a lateral air curtain model to improve cold conservation in cargo trucks during unloading.

Publications

- 09/2026 **Entanglement Certification by Measuring Nonlocality**, *Xuan Du Trinh*, Zhengyu Wu, Junlin Bai, Huan-Hsin Tseng, Nengkun Yu, Aruna Balasubramanian, *IEEE International Conference on Quantum Computing and Engineering (IEEE Quantum Week, QCE 2026)*, QNET track. [arXiv:2507.18066 \[quant-ph\]](#)
Nonlocality-based entanglement certification with applications to quantum networks. Lead author and presenter.
- 09/2026 **Emergence of Classical Stochasticity**, *Xuan Du Trinh*, Ismaël Septembre, Hai-Chau Nguyen, *Physical Review A*, accepted. [DOI:10.1103/855f-2529](#). Preprint: [arXiv:2602.04633 \[quant-ph\]](#)
Studies how classical stochasticity can emerge in quantum systems governed by Pauli-type master equations. Shows that stochastic times such as persistence and first-arrival times require an additional definite-state assumption, illustrated in the ultradecoherence limit for single particles, bosons, and fermions.
- 09/2026 **Full-Rank Noise Forbids Long-Range Entanglement Swapping**, *Xuan Du Trinh*, Nicholas Pardave, Angie Huang, Xiangyi Meng, Nengkun Yu, [arXiv:2609.11021 \[quant-ph\]](#)
Proves that any entangled two-qubit link with full-rank noise has a finite depth beyond which entanglement swapping can no longer establish end-to-end entanglement, even under adaptive postselection, while rank-deficient links keep a positive concurrence branch at every depth, with swapping experiments on a superconducting processor.
- 09/2026 **One Relative Phase Orders Operational Thresholds of Noisy Bell Pairs**, *Xuan Du Trinh*, [arXiv:2609.10034 \[quant-ph\]](#)
Shows that the X part of the noise alone bounds the Bell weight at which a noisy mixture gains an operational ability such as steerability or Bell nonlocality, so partial tomography can certify the ability, and proves that the threshold is nondecreasing in the relative phase between the noise and Bell coherences even when no closed-form criterion is known.
- 08/2026 **Operational Thresholds of Bell Mixtures with Complex X Noise**, *Xuan Du Trinh*, [arXiv:2608.17609 \[quant-ph\]](#)
Derives exact intervals and thresholds for entanglement, teleportation usefulness, steerability, and CJWR and CHSH violation in mixtures of a Bell state with arbitrary complex X noise, and shows the thresholds form a universal chain that widens as the relative phase between the noise and Bell coherences grows.
- 06/2026 **Rethinking Quantum Network Design Using a Verification-Based Quantum Transmission Protocol**, Yiming Zeng, Zhengyu Wu, *Xuan Du Trinh*, Yuanyuan Yang, Nengkun Yu, Aruna Balasubramanian, *46th IEEE International Conference on Distributed Computing Systems (ICDCS 2026)*. **Distinguished Paper Award**
- 06/2026 **Absorption Capacity of Separable Noise: Bell-Mixing Thresholds on Entanglement and Teleportation**, *Xuan Du Trinh*, [arXiv:2606.31243 \[quant-ph\]](#)
Defines the entanglement and fidelity absorption capacities of a Bell-mixing line, with closed-form thresholds for product and X noise states and extensions to local amplitude-damping and dephasing channels.
- 01/2026 **Manjushri: A Tool for Equivalence Checking of Quantum Circuits**, *Xuan Du Trinh*, Meghana Sistla, Nengkun Yu, Thomas Reps, [arXiv:2601.22372 \[quant-ph\]](#)
Scalable equivalence checking combining local-projection fingerprints with weighted binary decision diagrams, benchmarked on randomized Clifford+T circuits.
- 10/2025 **Scalable Equivalence Checking and Verification of Shallow Quantum Circuits**, Nengkun Yu, *Xuan Du Trinh*, Thomas Reps, *Proceedings of the ACM on Programming Languages*, Vol. 9, OOPSLA2
Scalable algorithms for verifying optimized quantum circuits produced by quantum compilers. Preprint: [arXiv:2504.01558](#) (04/2025).

09/2025 **Adaptivity Is Not Helpful for Pauli Channel Learning**, *Xuan Du Trinh, Nengkun Yu*, [Quantum](#), 9, 1864

Shows that adaptive strategies do not reduce the sample complexity of Pauli channel learning. Preprint: arXiv:2403.09033 (02/2024).

Talks

09/2026 **Entanglement Certification by Measuring Nonlocality**

IEEE Quantum Week (QCE 2026), QNET track, Toronto, Canada

10/2025 **Scalable Equivalence Checking and Verification of Shallow Quantum Circuits**

ACM SPLASH / OOPSLA 2025, Singapore

Teaching

Fall 2025 **Teaching Assistant, Analysis of Algorithms (graduate)**

Stony Brook University. Algorithms and data structures. Prepared homework and exam materials, held weekly office hours, graded assignments, and helped students with problem solving, debugging, and exam preparation.

Fall 2023 **Teaching Assistant, Foundations of Computer Science (undergraduate)**

Stony Brook University. Discrete mathematics and proof techniques. Prepared homework and exam materials, held office hours, graded assignments, and helped students with problem solving and exam preparation.

Mentoring

Summer 2025 **Research Mentor, High School Student**

Mentored a high school student through a multi-month research project, with weekly guidance on research planning, writing, and technical communication. The student was selected as a **Top 300 Scholar** in the **85th Regeneron Science Talent Search** (2,612 entrants), was admitted to MIT and Harvard, and will attend MIT.

Service

06/2024 **Reviewer**, *44th IEEE International Conference on Distributed Computing Systems (ICDCS 2024)*

02/2024 **Reviewer**, *International Joint Conference on Neural Networks (IJCNN 2024)*, *Quantum Machine Learning Track*

Honors and Awards

2026 **Distinguished Paper Award**

46th IEEE International Conference on Distributed Computing Systems (ICDCS 2026)

09/2023 – **Chairman's Fellowship**

09/2025 Department of Computer Science, Stony Brook University

10/2021 – **VALLET Scholarship**

09/2022 VALLET Foundation, France

09/2018 – **EIFFEL Scholarship**

08/2021 French Government, for outstanding international students

04/2016 **Third Prize**, Vietnamese Mathematical Olympiad for Undergraduate Students

03/2015 **Third Prize**, Vietnamese Physics Olympiad for High School Students

Workshops

06/2022 **Quantum Information and the Frontiers of Quantum Theory**

ENS de Lyon, France. Spacetime and quantum theory, observer physics and emergence of classicality, generalized probabilistic theories.

Skills

Programming Python, C++, C, Java, Julia, MATLAB, Bash, SQL

Libraries NumPy, SciPy, Pandas, Matplotlib, Qiskit, Quasimodo

Systems	Linux (Ubuntu, Debian) as daily environment, shell, SSH and remote development on shared servers, Git/GitHub, C++ build systems, <code>gdb</code> , <code>valgrind</code> , profiling, virtual environments
Data	Data processing and cleaning, statistical analysis, benchmarking pipelines, SQL (IBM Db2)
Scientific software	Mathematica, ROOT, COMSOL Multiphysics, Ansys Fluent
Quantum	Circuit simulation and verification, equivalence checking, channel learning, Pauli-noise characterization, entanglement and nonlocality certification, IBM Quantum hardware experiments
Mathematics	Probability and statistics, numerical linear algebra, linear and semidefinite programming, convex and numerical optimization, algorithms, Lie groups and representations
Physics	Quantum information, quantum optics, photonic and measurement-based quantum computing, open quantum systems, statistical physics, quantum field theory, many-body physics, general relativity, elementary particle physics
Other coursework	Natural language processing (neural networks, data preprocessing, model evaluation), database systems, statistics, programming languages, system security
IT	Windows, macOS, and Linux installation, configuration, and troubleshooting, basic networking (TCP/IP, DNS, VPN), printer and peripheral setup
Tools	LaTeX, Markdown, Microsoft Office, Google Workspace, Zoom, Teams, AI assistants (Claude, ChatGPT) for research, coding, and writing
Languages	Vietnamese (Native), English (Proficient), French (Proficient)